House Price Prediction – Findings Summary

This project aimed to analyse housing data and develop machine learning models capable of predicting residential property prices. The dataset contained information for 545 houses, including features such as area, number of bedrooms, bathrooms, stories, parking availability, air conditioning, basement presence, preferred area status, and furnishing condition. The objective was to identify the key factors influencing house prices and evaluate how accurately machine learning algorithms could estimate property values.

The dataset was first cleaned and prepared for analysis. No missing values or duplicate records were found, making the dataset suitable for modelling without extensive preprocessing. Categorical variables were converted into numerical form using binary encoding and one-hot encoding techniques. Additional exploratory analysis was performed through visualizations, including a price distribution histogram and a correlation heatmap, to understand relationships between features and the target variable.

Several features showed a strong relationship with house prices. Property area emerged as the most influential factor, followed by the number of bathrooms and the presence of air conditioning. The correlation analysis indicated that larger homes generally command higher prices, while premium amenities such as air conditioning and preferred locations also contribute significantly to property value. Feature importance analysis further confirmed that area was the dominant predictor of house price.

To predict housing prices, both Linear Regression and Random Forest Regression models were trained and evaluated. Linear Regression achieved the best overall performance, with an R^2 score of approximately 0.65, indicating that the model was able to explain around 65% of the variation in house prices. The model also achieved a mean absolute error of approximately ₹988,000, meaning predictions were typically within about one million rupees of the actual property value. Random Forest Regression achieved a slightly lower R^2 score of approximately 0.62 and did not outperform the simpler linear model.

Additional feature engineering techniques were explored by creating new variables such as area per bedroom, bathroom-to-bedroom ratio, a utility score representing premium amenities, and a luxury home indicator. While these engineered features provided additional insight into housing characteristics, they produced only marginal improvements in predictive performance. This suggests that the original dataset already captured most of the available information relevant to price prediction.

Overall, the project demonstrates that machine learning can effectively estimate housing prices using readily available property features. The analysis highlights that property size, number of bathrooms, and premium amenities are the strongest drivers of house value. Although the models achieved reasonable predictive accuracy, further improvements would likely require additional information such as neighbourhood quality, location coordinates, property age, and proximity to important facilities. The findings provide useful insights for homeowners, buyers, and real estate professionals seeking to understand the factors that influence housing prices.